Chapter 4: Incomplete Information and Signaling

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Outline

1. Introduction and Motivation
2. Adverse Selection vs. Moral Hazard
3. Bayesian Games and Equilibrium
4. Signaling Games
5. Brief Mention: Screening
6. Exercises and Wrap-Up

Why Study Incomplete Information?

- Many real-world interactions feature **asymmetric** or **incomplete** information:
  - A seller knows product quality better than buyers (adverse selection).
  - A job candidate knows his own ability better than employers (signaling).
  - Insurance buyers know their risk type better than insurers (screening).
- Traditional micro models often assume complete info, but that is rarely realistic.
- We wish to see how markets function (or fail) under information asymmetry and how to design mechanisms or signals to alleviate it.

Key Themes

- Bayesian games and Bayesian Nash equilibrium (BNE).
- Signaling and screening as ways to reduce adverse selection.
- Auctions, moral hazard, mechanism design.

Roadmap for Chapter 4

1. **Preliminaries: Adverse Selection vs. Moral Hazard**
2. **Bayesian Games**
   - Definition, BNE
   - First-Price Auction Example
3. **Signaling Games (Focus)**
   - A numeric example
   - Perfect Bayesian Equilibrium
   - On-/Off-path beliefs, the need for refinement
   - The intuitive criterion
4. **Brief Mention of Screening**
5. **Exercises & Wrap-Up**

Adverse Selection vs. Moral Hazard

Adverse Selection (Hidden Type)

- One side has private info about intrinsic characteristics (e.g. quality, risk type). - E.g. "Lemons problem" in used cars, high-risk drivers in insurance.

Moral Hazard (Hidden Action)

- One side's unobserved action affects outcomes (worker effort, driver's caution). - E.g. an insured driver might drive less carefully once insured.

Example (Insurance Setting)

- Adverse Selection: Type (risk) unknown to insurer. - Moral Hazard: After insured, driver exerts less caution.

Hidden Characteristics & Hidden Actions

- **Hidden Characteristics**: Private info about traits ex ante (borrower's risk, car's quality).
- **Hidden Actions**: Player's effort or behavior not observable ex post (employee effort).

Key Challenge

Design mechanisms (contracts, signals, or screening) so that private info or unobservable actions do not cause severe market failure (e.g. no trade, or inefficient coverage).

Classic Example: Market for Lemons (Akerlof 1970)

- Used cars with two qualities: high or low.
- Sellers know true quality, buyers only know distribution.
- If only one price, high-quality owners exit the market $\rightarrow$ only lemons remain.
- Potentially no trade or "lemons only" if no credible signal of quality.

Insight

Adverse selection can unravel markets unless there's a way to convey credible info (warranties, certification, etc.).

Bayesian Game: Setup

- A **Bayesian (normal-form) game** is specified by:
  1. Players $1, \ldots, n$.
  2. Type sets T_i for each player, with a **common prior** $\Pr(t_1, \ldots, t_n)$.
  3. Action sets A_i (or pure strategies).
  4. Payoffs $u_i(a_1, \ldots, a_n; t_1, \ldots, t_n)$.
- Each player knows its own type t_i but not others' t_j.
- They form beliefs about others' types using the **common prior** *conditional on their own type*.

Strategy in a Bayesian Game

- A **(pure) strategy** s_i is a function from your type t_i to an action a_i:

$$s_i : T_i \rightarrow A_i.$$

- In deciding how to act for each t_i, a player **anticipates** other players' strategies $s_{-i}(\cdot)$ and forms beliefs about their types from the prior + own type.

Clarification

"Anticipating" means each player forms an expectation about how others will *behave*, not that they assume "others choose the same as me." They use the equilibrium concept to guess s^*_{-i}.

Bayesian Nash Equilibrium: Definition

Idea: Each player maximizes expected payoff given their type, using the **common prior** updated by their own type, plus knowledge of others' strategies.

Definition (Bayesian Nash Equilibrium)

A strategy profile $(s_1^*, \ldots, s_n^*)$ is a **BNE** if $\forall i,\ \forall t_i$,

$$s_i^*(t_i) \in \arg\max_{a_i \in A_i} \mathbb{E}\Big[u_i\big(a_i, s_{-i}^*(t_{-i});\ t_i, t_{-i}\big) \,\Big|\, t_i\Big].$$

Remark

- **Beliefs** *about t_{-i} come from the common prior, conditioned on t_i. - The other players' strategies affect payoffs, but not the prior about types.*

Tiny Bayesian Game (Example)

Players: 2. **Types**: Each player has 2 possible types, $T_i = \{\alpha, \beta\}$.
Actions: $A_i = \{L, R\}$ for each.
Common prior: $\Pr(t_i = \alpha) = 0.5, \Pr(t_i = \beta) = 0.5$, independent across players.

Example (Payoffs)

- If both choose L, payoffs depend on whether type is $(\alpha, \alpha), (\alpha, \beta), \ldots$. *- Suppose* $u_1(L, R; \alpha, \beta) = 3$ *etc. (omitted details).*

Find BNE

We define strategies $s_i(\alpha) \in \{L, R\}, s_i(\beta) \in \{L, R\}$, then check best responses in expectation over the other's type.

First-Price Sealed-Bid Auction (Private Values)

Setup:

- n bidders, each has private valuation $v_i \in [0, 1]$ for the item, i.i.d. with cdf F.
- Bids are sealed b_i. Highest bidder wins, pays own bid.
- **Type**: $t_i = v_i$. **Action**: $b_i = s_i(v_i)$.

Claim

In symmetric equilibrium with i.i.d. uniform(0,1), each uses a bidding function $b^*(v) = \frac{n-1}{n} v$.

Remark

We'll illustrate the idea for $n = 2$ to see how the equilibrium is derived, then generalize.

First-Price Auction Derivation (n=2, Uniform[0,1])

Assume a symmetric BNE: $b^*(v)$.
Focus on player 1 with valuation v_1:

- If player 2 uses $b^*(v_2) = \beta\, v_2$ for some $\beta < 1$,
- Then player 1 chooses b_1 to maximize expected utility:

$$U_1 = (v_1 - b_1)\ \Pr(\text{win}).$$

- Probability of winning $= \Pr(b_1 > b^*(v_2)) = \Pr\big(b_1 > \beta v_2\big)$.

First-Price Auction Derivation (n=2, cont.)

For uniform $v_2 \in [0,1]$,

$$\Pr(\text{win}) = \Pr(\beta v_2 < b_1) = \Pr\left(v_2 < \frac{b_1}{\beta}\right) = \frac{b_1}{\beta}, \quad \text{provided } b_1/\beta \leq 1.$$

Hence,

$$U_1 = (v_1 - b_1)\,\frac{b_1}{\beta},$$

for $b_1 \leq \beta$ (since $b_1/\beta \leq 1$).
Take derivative wrt b_1:

$$\frac{dU_1}{db_1} = \frac{1}{\beta}\big[v_1 - 2b_1\big].$$

Setting $= 0$, we get $b_1^* = \frac{v_1}{2}$, i.e. $\beta^* = \frac{1}{2}$.

Result

In BNE for $n = 2$, $b^*(v_i) = \frac{1}{2}v_i$. For general n, it's $\frac{n-1}{n}v_i$.

BNE vs. PBE: What's the Difference?

- **Bayesian Nash Equilibrium (BNE)**: A solution concept for *simultaneous-move* (normal-form) games of incomplete info. Players choose strategies mapping "type $\rightarrow$ action."
- **Perfect Bayesian Equilibrium (PBE)**: A refinement concept typically for *dynamic or sequential* games with incomplete info.
  - Adds **beliefs at each information set** (including off-path).
  - Requires **sequential rationality** + Bayes' rule to update beliefs on each possible path.

Summary

BNE is for **static** incomplete info. PBE extends to **dynamic** settings, ensuring rationality at every node (subgame) with consistent beliefs.

Signaling Game Setup

- Typically a **dynamic** game:
  1. Nature draws Sender's type t (hidden from the Receiver).
  2. Sender observes t, chooses a signal s (e.g. invests in education).
  3. Receiver observes s but not t, then chooses an action a.
- Payoffs: $u_S(t, s, a)$, $u_R(t, s, a)$.
- We want a solution concept: **Perfect Bayesian Equilibrium (PBE)**, which imposes beliefs $\mu(t \mid s)$ and sequential rationality.

Remark

BNE is for simultaneous incomplete-info games. **PBE** *is more refined for sequential or extensive-form settings, requiring "on-path" beliefs from Bayes, and specifying "off-path" beliefs.*

On-Path vs. Off-Path Beliefs

- **On-path signals**: The signals that arise with positive probability in equilibrium.
- **Off-path signals**: Signals that do not occur (or occur with probability 0) in equilibrium.

Belief Updating

- **On-path**: Use Bayes' rule to update $\mu(t \mid s)$ from prior and equilibrium strategy. - **Off-path**: By definition, s occurs with prob 0 under the equilibrium strategy. Bayes' rule is not defined (0 in the denominator), so the game can specify beliefs in some "reasonable" manner.

Remark

Refinement concepts often revolve around restricting how players assign beliefs to off-path signals, ruling out "unreasonable" equilibria.

A Simple Numeric Example

Basic story:

- Worker type $t \in \{\text{High}, \text{Low}\}$, each with probability 0.5.
- Ability: H yields productivity 10, L yields 0.
- Employer pays wage = worker's "believed ability." (So if believed H, pay 10; if believed L, pay 0; or if uncertain, pay the average.)
- Signal $s \geq 0$ represents "units of education."
- Cost of s for H is s, for L is $2s$.

Payoff to Worker

Worker payoff = wage − cost(s, type).
Off-path beliefs about s-values not used in equilibrium will be crucial to support or rule out certain outcomes.

Potential Equilibria (From the Example Statement)

- **Separating** equilibria with $s_H^* \in [5, 10]$ and $s_L^* = 0$:
  - If H invests e.g. $s_H^* = 6$, the firm sees $s = 6$ and believes "this must be High," paying wage 10.
  - H payoff $= 10 - 6 = 4$. L chooses 0, is believed to be Low, so wage=0, payoff $= 0 - 0 = 0$.
  - We must ensure L does not deviate to some $s > 0$ in hope of being paid 10. This depends on *off-path* beliefs and cost differences.
- **Pooling** equilibria with $s^* \in [0, 2.5]$:
  - Both types pick the same s^*, so the firm pays the average ability of 5.
  - We must ensure no incentive for H to deviate upward and become recognized as High. This also depends on how the firm reacts off-path.

Why Do These Equilibria "Hold"? (Off-Path Beliefs)

- In a **separating** equilibrium, if the firm *always* believes that a worker who invests $s \geq s_H^*$ is High, then Low type cannot profitably mimic (since cost $= 2s$, which might exceed the wage gain).
- In a **pooling** equilibrium, if the firm's *off-path* belief says "any worker deviating to a slightly higher s is probably Low," the deviant might not receive a higher wage, thus no payoff gain for H.
- Equilibrium or not depends heavily on how we assign beliefs for signals not used on the path (the *off-path signals*).

Important

Hence, to show an equilibrium is stable, we must specify how the firm forms its belief $\mu(\text{High} \mid s)$ for each possible s, including those not chosen in equilibrium. That can prevent beneficial deviations.

Intuitive Criterion: Ruling Out Unreasonable Equilibria

- If there is a certain off-path deviation s' such that:
  1. Under the *best possible belief* for the deviator (namely, the firm believes he is High), the High type's payoff strictly increases if it chooses s'.
  2. Even under that same optimistic belief, the Low type's payoff would *decrease* if it chooses s'.
- Then the firm *should* interpret any worker who actually does s' as High. This eliminates any equilibrium that relied on some contradictory off-path belief about s' (like "$s' = 5.1$ must be Low").

Core Logic

High type can deviate profitably if the firm is "kind" enough to interpret it as High. But for Low type, even that "kind" belief is not beneficial. So only H would want to deviate there.

Example: Why $s^* = 6$ for H Might Be Ruled Out

Given the continuum of separating equilibria, suppose in one equilibrium, H invests $s^* = 6$, L invests 0.

- H payoff $= 10 - 6 = 4$.
- L payoff $= 0 - 0 = 0$.

Off-path deviation: $s' = 5.1$.

- If the firm sees $s = 5.1$ and *believes* "this must be High," it pays 10. - Then H's payoff would be $10 - 5.1 = 4.9 > 4$, better than 6. - L's payoff would be $10 - 2(5.1) = -0.2 < 0$, so L would not want that deviation.

Hence the *intuitive criterion* says the firm should interpret $s = 5.1$ as High only, thus H can profitably deviate from 6 down to 5.1. So the $s^* = 6$ separating equilibrium is not stable.

Example: Why All Pooling Equilibria Fail

A pooling eq might be $s^* = 0$ for both H and L, so the wage is 5 (the average ability).
Deviation $s' = 4$:

- If the firm believes that any worker who invests 4 is High, it will pay 10.
- Then H payoff is $10 - 4 = 6$ (up from $5 - 0 = 5$). So H would be *better off* by deviating to 4 *if* the firm interprets him as High.
- L payoff if believed High would be $10 - 2(4) = 2$ (down from $5 - 0 = 5$). So L would *not* deviate.

So that $s' = 4$ deviation can only come from High type. The intuitive criterion says the firm should treat that deviation as High. Then H gains, which kills the $s^* = 0$ pooling eq.

PBE: Beliefs + Sequential Rationality

- **Strategies**:
  - Sender: which s to choose for each type t.
  - Receiver: after observing s, choose an action $a(s)$.
- **Belief system** $\mu(t \mid s)$ for each possible s (on- or off-path).
- **Conditions**:
  1. Bayes rule on-path.
  2. Sequential rationality: given beliefs, the Receiver's action is optimal. Given the Receiver's strategy, the Sender's choice for each type is optimal.
  3. Possibly a refinement like the *intuitive criterion* to discard "unreasonable" off-path beliefs.

Summarizing Signaling

- **High type** can separate from Low type by investing in a "signal" that is costlier for L than for H.
- **Multiple equilibria** can arise: various separating s^* or pooling.
- **Off-path beliefs** matter to show no profitable deviation.
- The **intuitive criterion** (Cho-Kreps) or other refinements can rule out many seemingly "bizarre" equilibria.

Practical Examples

- Education (Spence model) - Warranties, brand advertising, etc.

Screening: A Very Short Mention

- **Screening** is the flip side of signaling, where the *uninformed* side (e.g. an employer or an insurer) designs a **menu** of contracts or prices to induce self-selection.
- Example: Different coverage/premium plans in insurance for high-risk vs. low-risk types, or multiple product versions for high-valuation vs. low-valuation consumers.

We'll Defer Details

In a subsequent chapter on **mechanism design**, we will delve into screening more thoroughly (Rothschild-Stiglitz insurance model, menus of contracts, etc.).

Adverse Selection: Summarizing the Tools

Scenario: One side has private info ex ante (type).

- **Signaling (Sender-Initiated)**: The informed party invests in a costly action that credibly reveals type.
- **Screening (Receiver-Initiated)**: The uninformed party designs a menu of choices to induce self-selection.
- If neither side can effectively signal/screen, the market may unravel (lemons problem).

Policy Angle

Governments sometimes intervene (e.g., mandatory disclosure, minimum standards, consumer protection) when private signaling/screening is insufficient or too costly.

Exercises (1)

1. **Market for Lemons**:
   - Suppose fraction α of cars are high quality (v_H) and fraction $1-\alpha$ are low quality (v_L) with $v_H > v_L$.
   - Show how adverse selection can cause only lemons to trade if there's no credible signal of quality.
2. **Bayesian Game Basics**:
   - Construct a 2-player Bayesian game where each player has 2 possible types. Write out the payoff matrix for each type profile.
   - Propose strategies and check for a Bayesian Nash Equilibrium.

Exercises (2)

1. **First-Price Auction Equilibrium**:
   - Two bidders, valuations v_1, v_2 i.i.d. Uniform(0,1).
   - Show (or sketch) that the BNE is $b_i(v_i) = \frac{n-1}{n} v_i$ for n bidders.
2. **Signaling Game Setup**:
   - Write a simple 2-type, 2-signal game (sender chooses $s \in \{s_1, s_2\}$, receiver chooses $a \in \{a_1, a_2\}$).
   - Solve for possible pooling or separating equilibria.
   - Identify conditions for each type of equilibrium.

Exercises (3)

1. **Spence Model Variation**:
   - Suppose $C_H(e) = e$, $C_L(e) = 2e$, productivities are $\theta_H = 10, \theta_L = 5$.
   - Find a separating equilibrium $e_H > e_L$. What wages are offered? Check incentive constraints.
   - Illustrate a scenario in which the *Intuitive Criterion Cho-Kreps* might rule it out.
2. **Adverse Selection vs. Moral Hazard**:
   - Provide a real-world example where both adverse selection and moral hazard arise.
   - Suggest policy or contractual tools to address each problem distinctly.

References

- Akerlof, G. *"The Market for Lemons,"* (1970).
- Spence, M. *"Job Market Signaling,"* (1973).
- Mas-Colell, Whinston, Green. *Microeconomic Theory*, Ch. 13–14.
- Fudenberg & Tirole. *Game Theory*, sections on Signaling and Refinements.

Concluding Remarks

- Incomplete information fundamentally alters market outcomes compared to full-information models.
- **Bayesian games** provide a framework for simultaneous incomplete-info interactions (like auctions).
- **Signaling games** cover sequential moves where the informed side invests in a costly signal. Off-path beliefs are crucial; refinements (intuitive criterion) prune odd equilibria.
- **Screening** is briefly introduced, with more to come in mechanism design.

Q & A

Thank you for your attention!